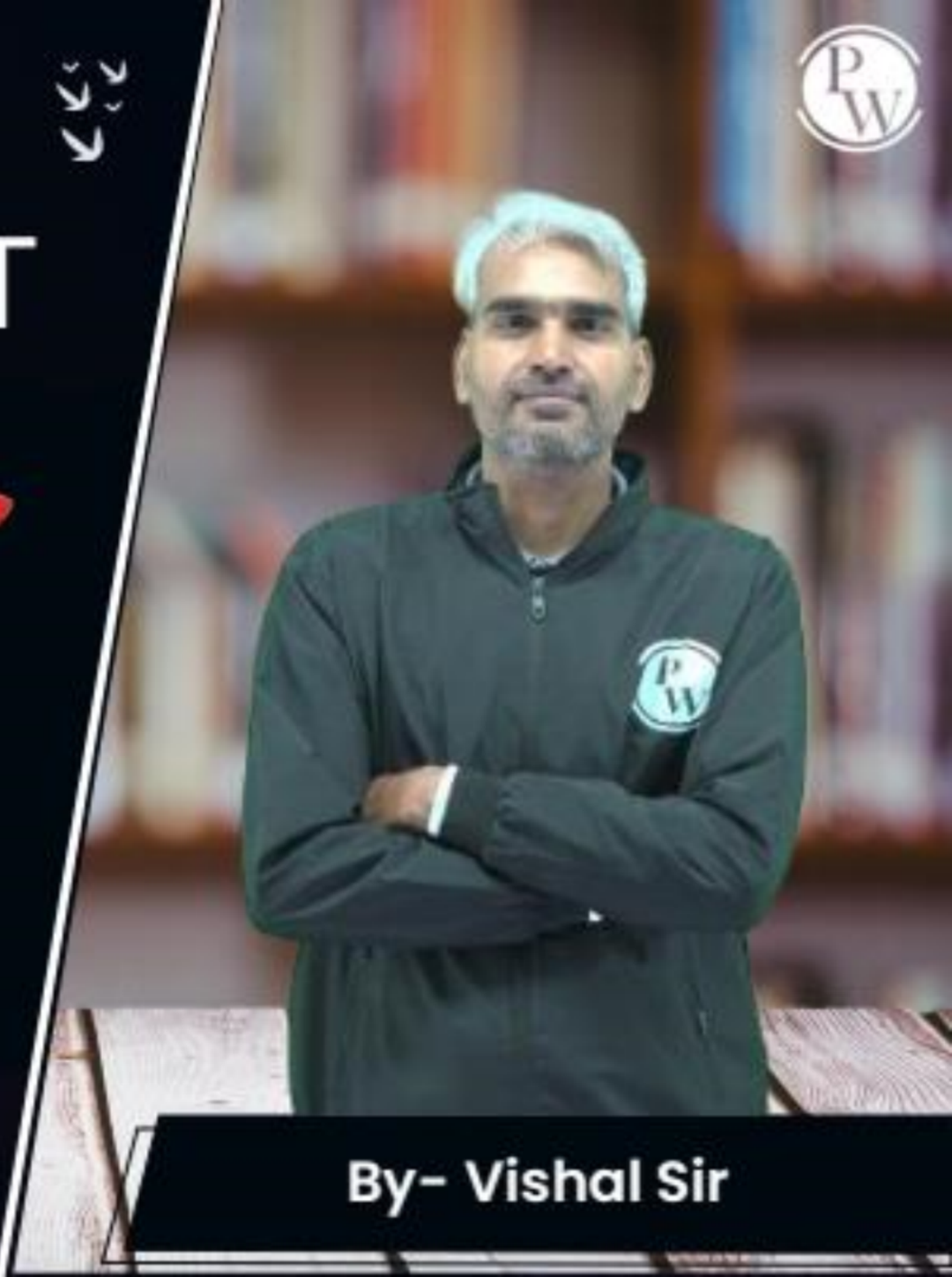
Computer Science & IT



Discrete Mathematics

Graph Theory

Lecture No. 06



By- Vishal Sir

Recap of Previous Lecture



- Topic: Graph isomorphism
- Topic: Self-complementary graph
- Topic: Planar graph

In a Self-complementary graph G_1.

$$|E(G_1)| = \frac{n(n-1)}{4}$$

$$|V| = 4k \text{ or } (4k+1)$$

Slide

Topics to be Covered

Topic	Planar graph



Slide



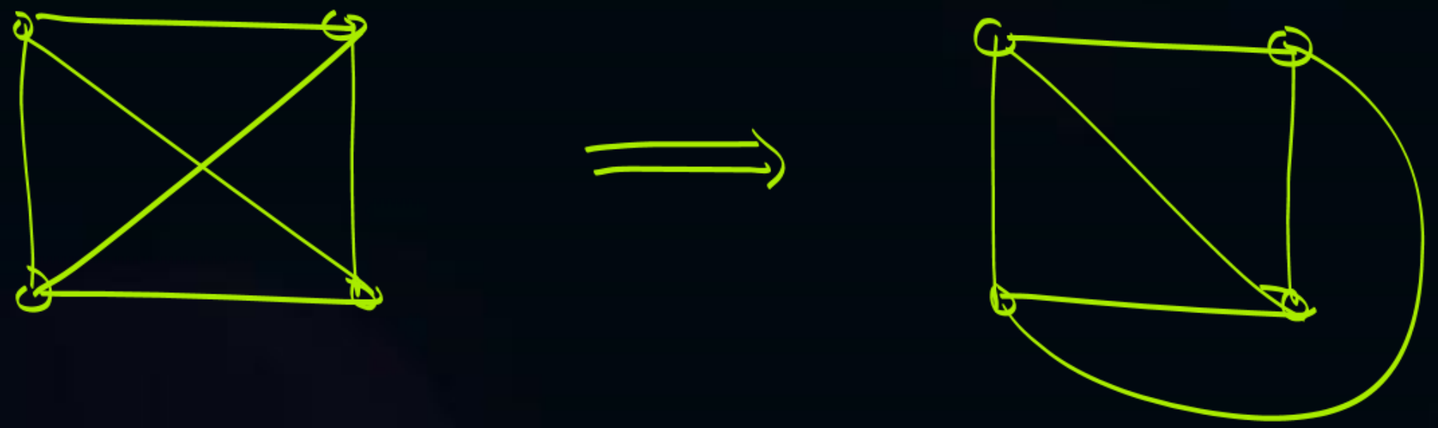
Topic : Planar graph

A graph G= (V, E) is said to be planar if it can be drawn in the plane such that no two edges of G intersect each other at a non-vertex point.

Such a drawing of a planar graph is called a planar embedding of the graph.



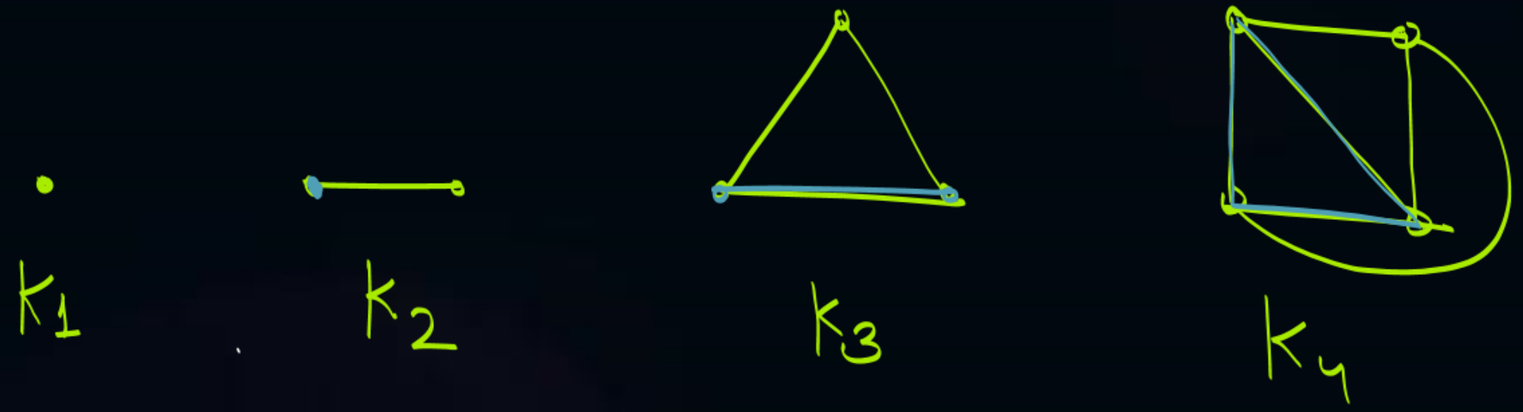
Topic : Planar graph



Planar graph



Topic : Planar graph



K_n is planar if and only if $n \leq 4$

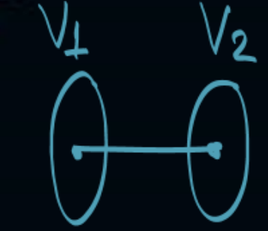


K_5

K_5 is not Planar

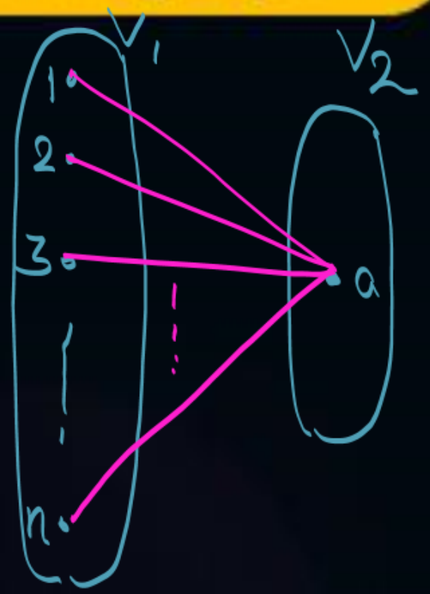


Topic : Planar graph



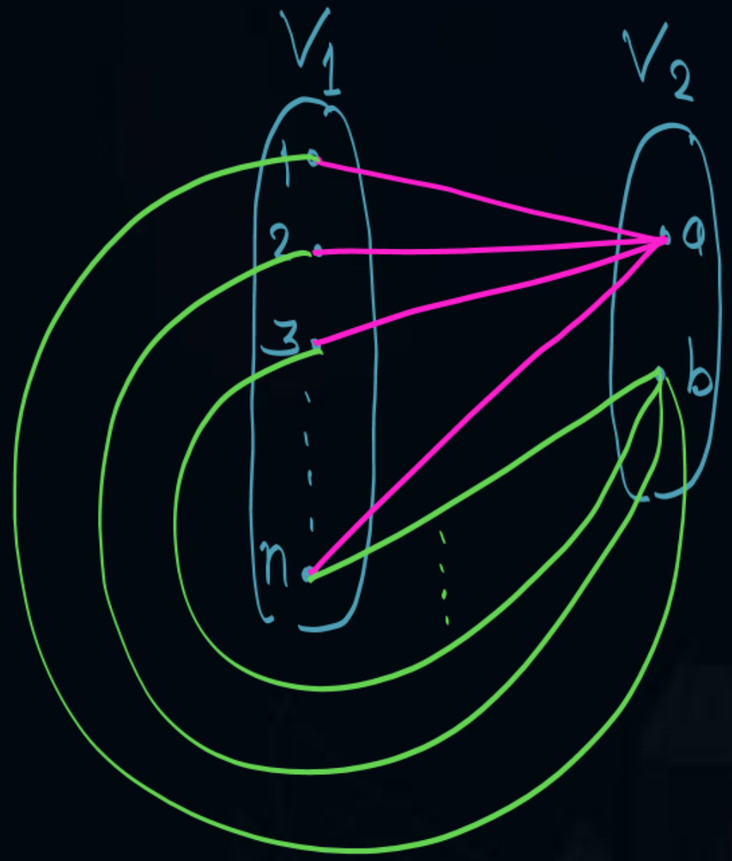
$K_{1,1}$

{Planar}



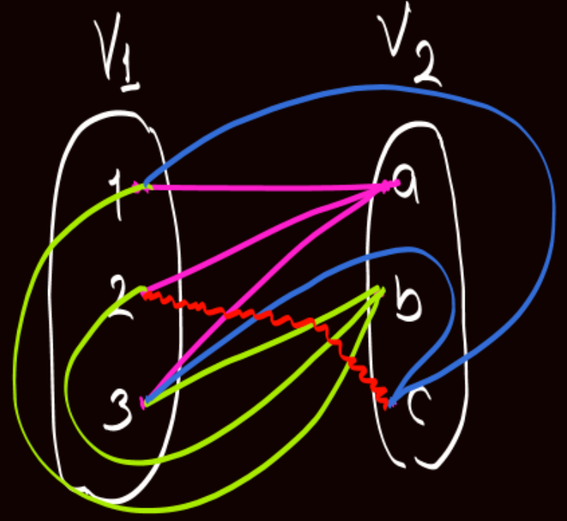
$K_{n,1}$ is planar

Similarly $K_{1,n}$ is also planar



$K_{n,2}$ is planar

Similarly $K_{2,n}$ is also planar



$K_{3,3}$ is not planar

Note: A complete bipartite graph $K_{m,n}$ is planar if and only if $m \leq 2$ or $n \leq 2$



Topic : Planar graph

* K_5 is a non-planar graph with 5-vertices & 10-edges
* $K_{3,3}$ is a non-planar graph with 6-vertices & 9-edges

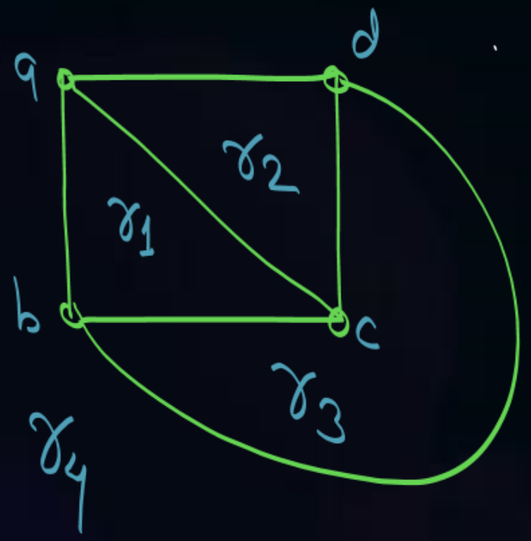
→ Non-planar graph with minimum number of vertices is K_5

• Non-planar graph with minimum number of edges is $K_{3,3}$

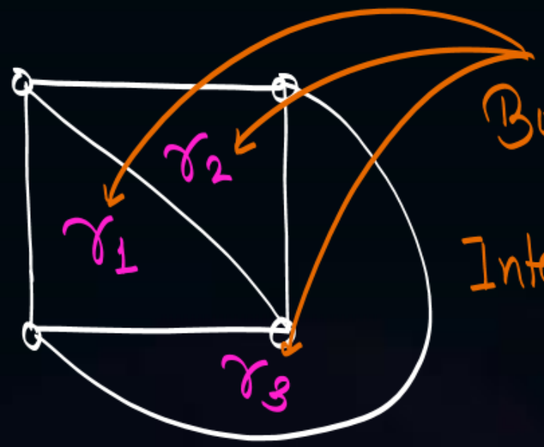


Topic : Region / Face

Every planar graph divides a plane into connected areas called as regions of the plane or faces of the planar graph.

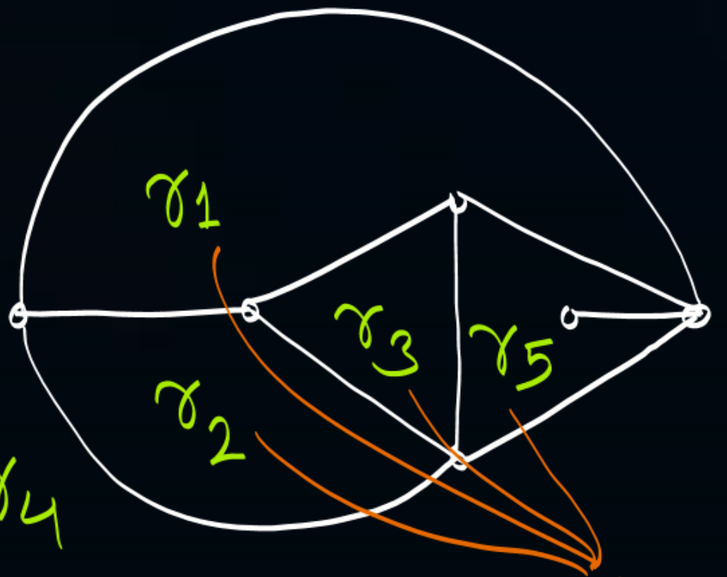


Topic : Region / Face



Bounded region
(or)
Interior regions

Unbounded region
(or)
Exterior region



Exterior region
(Unbounded region)

Interior regions
(Bounded region)



Topic : Region / Face

Note: In any planar graph there is exactly '1' unbounded region, all other regions are called bounded regions.

$$\therefore \text{No. of Bounded regions in a planar graph} = \left(\text{Total no. of regions in that planar graph} - 1\right)$$



Topic : Degree of a region/face

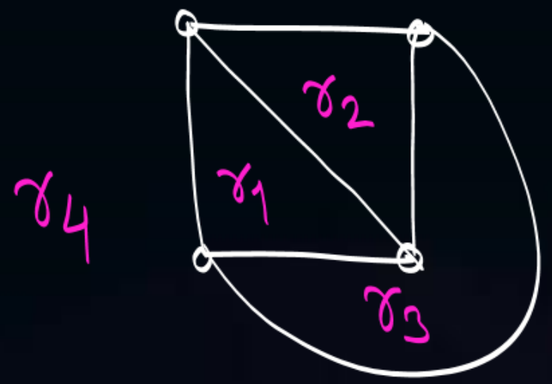
Degree of an interior region = No. of edges enclosing that region
(or)
Number of edges on the boundary of that region

Degree of Exterior region = Number of edges exposed to that region
(or)
Number of edges that seperate the region from the other regions of the graph

Note: Degree of a region 'r' is denoted by deg(r).



Topic : Degree of a region/face



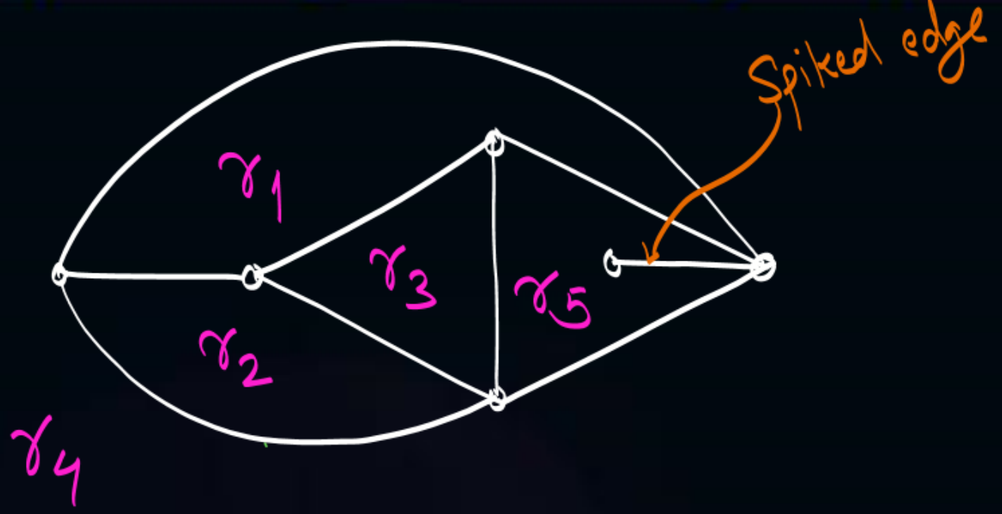
$\deg(r_1) = 3$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

Topic : Degree of a region/face



$\deg(r_1) = 4$

$\deg(r_2) = 3$

$\deg(r_3) = 3$

$\deg(r_4) = 3$

$\deg(r_5) = 3 + 2 = 5$

3 → Normal edges

2 → Spiked edge

Note:- Spiked edge in a region is counted as two edges in-order to obtain the degree of that region

Topic : Sum of degree theorem

{ wo.t degree of regions }



$|R| = 3$

$= \deg(r_1) + \deg(r_2) + \deg(r_3)$

$= 3 + 3 + 4$

$= 10 = 2 * 5$

No. of edges

Because of one edge overall increment in the degrees of regions of the graph is '2'

Let $|R|$ denotes the number of regions in the planar graph; then

$$\sum_{i=1}^{|R|} \deg(r_i) = 2|E|$$



Topic : Sum of degree theorem

Corollary 1:- No. of regions with odd degree must be even.



Topic : Sum of degree theorem

Corollary 2 :- In a planar graph if degree of each region is exactly 'k', then

$$\sum_{i=1}^{|R|} \deg(r_i) = 2|E|$$

$$\underbrace{k + k + \ldots k}_{|R| \text{ times } 'k'} = 2|E|$$

$$\boxed{k \cdot |R| = 2|E|}$$



Topic : Sum of degree theorem

Corollary 3 :- In a planar graph if degree of each region is at least 'k' {i.e. $\geq k$}, then then

$$k|R| \leq 2|E|$$



Topic : Sum of degree theorem

Corollary 4 :- In a planar graph if degree of each region is at most 'k' {i.e $\leq k$}, then then

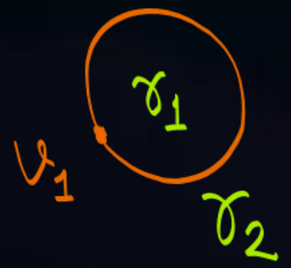
$$k|R| \geq 2|E|$$



Topic : Simple planar graph

{ A planar graph with no loop & no parallel edges is called a simple planar graph. }

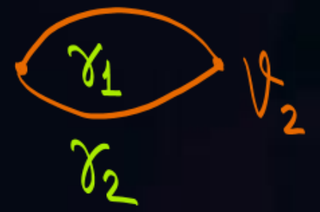
Loop:



"$\deg(r_1) = 1$"

{ degree of a region can be one only w.r.t. loop }

Parallel Edges:



"$\deg(r_1) = 2$"

{ degree of a region can be '2' only w.r.t. parallel edges }

i.e, In a simple planar graph degree of region can never be "1" or "2".



Topic : Simple planar graph

- In a simple planar graph with two or more regions degree of each region is at least "3"

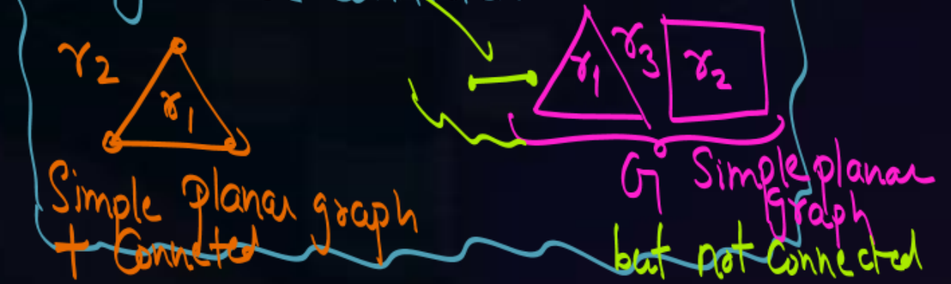
i.e. In a simple planar graph

$3|R| \leq 2|E|$ will always hold true

{ if degree of each region is at least 'k' then, $k.|R| \leq 2|E|$ }

$$3|R| \leq 2|E| \quad \text{---} \quad \text{eq}^{n} \ (1)$$

Simple Planar graph may or may not be connected



Simple Planar graph + Connected

G Simple planar Graph but not Connected



Topic : Euler's equation

(For Connected Planar graph)

→ May or may not be simple

For any Connected Planar graph,

$|V| + |R| = |E| + 2$ will hold true

Where,

$|V|$ = No. of vertices

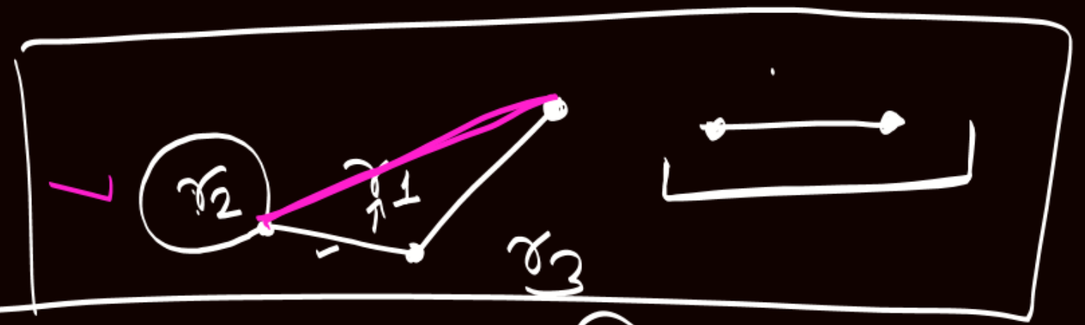
$|R|$ = No. of regions

$|E|$ = No. of Edges

let n = No. of vertices
e = No. of edges
f = No. of faces (regions)
then, $n + f = e + 2$
or $n - e + f = 2$

$$|V| + |R| = |E| + 2 \quad \text{— eqn (2)}$$

(Will not hold in a disconnected planar graph)



$|V| + |R| = |E| + 2$

$1 + 1 = 0 + 2$ ✓

$1 + (1+1) = 1 + 2$ ✓

$(1+1) + (1+1) = (1+1) + 2$

$(1+1+1) + (1+1) = (1+1+1) + 2$

$(1+1+1) + (1+1+1) = (1+1+1+1) + 2$



Topic : Simple connected planar graph

Simple Connected Planar graph = (Simple planar graph + Connected Planar graph)

Simple planar graph → $\therefore 3|R| \leq 2|E|$ — eq^{n} (1) will hold true.

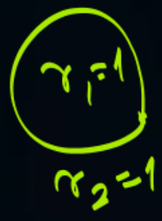
Connected Planar graph → $\therefore |V| + |R| = |E| + 2$ — eq^{n} (2) will also hold true

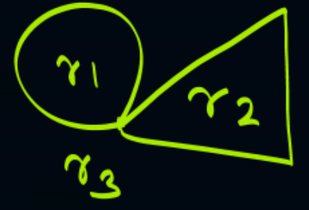
Note: ① In a simple planar graph eq^{n} ① will always hold true, but eq^{n} ② may or may not hold true because need not be connected

② In a connected planar graph eq^{n} ② will always hold true, but eq^{n} ① may or may not hold true because it need not be simple



Topic : Note





③ In a simple connected planar graph both eq^n ① as well as eq^n ② will always hold true

$3|R| \leq 2|E|$ — eq^n ① & $|V|+|R| = |E|+2$ — eq^n ②

By eq^n ① $|R| \leq \frac{2|E|}{3}$

Put this value of $|R|$ in eq^n ②

$$|V| + \frac{2|E|}{3} \geq |E|+2$$

$$|E| \leq 3|V|-6 \quad \text{— eq}^n \text{ ③}$$

$$e \leq 3n-6$$

By eq^n ① $|E| \geq \frac{3|R|}{2}$

Put this value of $|E|$ in eq^n ②

$$|V| + |R| \geq \frac{3|R|}{2} + 2$$

$$|V| \geq \frac{|R|}{2} + 2 \quad \text{or} \quad |R| \leq 2|V|-4 \quad \text{— eq}^n \text{ ④}$$



Topic : Note

④ In a simple connected planar graph all four equations eqⁿ ①, eqⁿ ②, eqⁿ ③ & eqⁿ ④ will always hold true.



2 mins Summary

Topic Planar graph


THANK - YOU
Slide